Md Anwarul Islam Aion

Avionics Engineering, Aviation and Aerospace University, Bangladesh (AAUB)
ResearchGate: Anwarul Islam Aion | Google Scholar: Anwarul Islam Aion
LinkedIn: anwarul-islam-aion | GitHub: aionndc22 | OrcID: 0009-0004-4061-0150
Web: anwarulislamaion.engineer | Email: aion.aaub@gmail.com | Contact: +8801717617918

Objective

Aviation has always inspired me to pursue engineering that creates a meaningful impact. Backed by a strong academic foundation in Avionics Engineering, hands-on aerospace engineering projects, multiple international research publications and conference presentations, leadership experience, and a passion for continuous learning, I am committed to applying my analytical thinking, technical expertise, and engineering mindset to advance safe, reliable, and innovative aviation systems.

Skills

Programming & AI	**Python** (NumPy, SciPy, scikit-learn, OpenCV), **C++**, Machine Learning, Deep Learning, Computer Vision (**YOLOv8**)
Modeling & Simulation	**MATLAB**/Simulink (Signal Processing & System Modeling), **GTSAM** (Factor-Graph Estimation), **CST Studio Suite** (RF & Antenna Design), **LTspice** (Analog & Power Circuit Simulation)
Hardware & Embedded Systems	**Altium Designer** (PCB Design), **FreeRTOS** (Embedded Real-Time Systems), **SDR** (ADALM-Pluto, GNU Radio), Real-Time **C++** (CMake, Ring Buffers, UDP Streaming)
Design & Development Tools	**Qt 6** (Desktop GUI Development), **React** (Vite, Tailwind CSS), Git & GitHub, pytest, Visual Studio Code
Technical Writing	**LaTeX** (Scientific Writing), Notion (Knowledge Management)
Leadership & Communication	Team Leadership, Technical Presentation, Proposal Writing, Fundraising

Education

BSc in Avionics Engineering 2023–2027 (Expected)
Aviation and Aerospace University, Bangladesh (AAUB) *Lalmonirhat, Bangladesh*

- CGPA: **3.62/4.00** (till 6th semester). (under the supervision of **Bangladesh Air Force**)
- *Undergraduate Thesis:* **RIS-assisted Integrated Sensing and Communication (ISAC)**, in two subsystems — the range-resolution limits of a reconfigurable intelligent surface, derived as an element-bandwidth criterion and tested against CST full-wave simulation and hardware measurement; and 5G downlink sensing, measuring how sensing degrades as scheduler load changes.
- *Relevant Coursework:* Fourier & Laplace Transform, Modeling and Simulation, Signal and System, Digital Signal Processing, Communication System, Microprocessor & Microcontroller, Electrical & Electronic Circuit Analysis, Data Structure & Algorithm, Electromagnetic Field Theory, Digital Logic & Computer Design, Real Time Embedded System, VLSI, Microwave & Antennas, Electric Machines & Drive, Control System, Avionics System Design, RADAR Engineering.

Higher Secondary Certificate (HSC) 2022
Notre Dame College; Result: ***Perfect GPA.*** *Dhaka, Bangladesh*

Secondary School Certificate (SSC) 2020
Kushtia Zilla School ; Result: ***Perfect GPA*** *with National Merit Scholarship.* *Kushtia, Bangladesh*

Professional Experience

Research Assistant 2026–Present
AAUB Radar Project *(Aviation and Aerospace University, Bangladesh)* *Lalmonirhat, Bangladesh*
Supervisor: Air Commodore Dr. Mohammad Akhter Jamil Hours per week: 20

- Appointed by special exception as the **first undergraduate in the university's history** to hold a paid research assistantship.
- Building a Pulse-Doppler radar: end-to-end DSP covering I/Q signal processing, MTI and CFAR detection; real-time data acquisition (DAQ) with UDP streaming; and legacy radar display integration, targeting an FPGA-compatible implementation.

Research Publications

[1] **The Cross-Track Information Cost of Weakly Observable Landmark Orientation: A Closed-Form Retention Law and Its Operating Conditions in Visual Teach-and-Repeat**. *Journal of Field Robotics* (Wiley), 2026. **Under review**.

[2] **Range-Resolution Limits of RIS-Assisted Radar: An Element-Bandwidth Criterion and Its Full-Wave Test**. *IEEE International Conference on Communications (ICC)*, SAC: Integrated Sensing and Communication, 2027. **Submitted**.

[3] **A Framework for Resilient Power Systems in CanSat-class Reusable Small Satellites**. *26th International Middle East Power Systems Conference (MEPCON)*, Egypt, 2025. DOI: 10.1109/MEPCON66918.2026.11360120.

[4] **LIFE-MAP: An Indigenous Biosignature Detection Protocol Using Multi-Biomolecule Based Analysis**. *IEEE Region 10 Conference (TENCON)*, Malaysia, 2025. DOI: 10.1109/TENCON66050.2025.11375044.

Engineering Projects

[1] **Team Lead** 2026–Present
Electrical Engineer 2024–2025
AAUB CanSat *[Team Ababil]* *Lalmonirhat, Rangpur, Bangladesh*

- Core electrical and power-systems engineer for a CanSat-class small satellite; contributed to the team achieving country's first **Global Finalist** at the CanSat 2025 finals in **Virginia, USA** (American Astronautical Society & NASA).
- Co-authored and presented a peer-reviewed research paper publication.

[2] **Science Module Lead** Nov 2024–Apr 2026
AAUB Mars Rover *[Team Rover71]* *Lalmonirhat, Rangpur, Bangladesh*

- Directed scientific mission planning and developed the Science & Environment Analysis subsystem from scratch for an indigenous Mars rover; coordinated subsystem integration with the EPS team, achieving **8th globally** in the Anatolian Rover Challenge 2025.
- Co-authored a peer-reviewed publication on the indigenous biosignature detection framework.

[3] **Team Lead** 2024–25
Oblique-Wing UAV Project *[Independent]* *Bangladesh*

- Designed and built an oblique-wing aircraft prototype inspired by the **NASA AD-1** configuration.

- Developed a custom 3D-printed EDF propulsion unit with thrust-configuration testing.

[4] **Case study** Nov 2024
The Technological Advancement of Electric Propulsion *Bangladesh*
- Presented on ion-thruster electric propulsion systems principle, market applications and future outlook at Impulse 1.0 at AAUB.

Industrial Attachment & Internships

214 MROU (BAF): F-7 Fighter Aircraft Overhaul & Avionics Systems 1 week
216 MROU (BAF): Mi-17/Mi-171 Helicopter Overhaul & Avionics Maintenance 1 week
208 MROU (BAF): Bell 206/212 Helicopter Overhaul & Electrical Systems 1 week
205 MU (BAF): Electronic Systems Maintenance, PCB Development & Test Engineering 1 week
133 ATU (BAF): C-130J Super Hercules Avionics & Electrical Systems 1 week
Biman Bangladesh Airlines: Boeing 787 Dreamliner Line Maintenance, CAMO, AMO & Quality Assurance 1 week

Leadership & Volunteering

President 2026–Present
AAUB Debating Club (AAUB-DC) *Lalmonirhat, Rangpur, Bangladesh*
Office Secretary 2022–2023
Kushtia Debating Society (KDS) *Kushtia, Bangladesh*
Core Debater & Executive Member 2021–2023
Notre Dame Debating Club (NDDC) *Dhaka, Bangladesh*
Vice President 2020–2022
Kushtia Science Research Club (KSRC) *Kushtia, Bangladesh*
Core Debater & Joint General Secretary 2018–2020
Kushtia Zilla School Debating Club (KZSDC) *Kushtia, Bangladesh*

Grants

Aviation and Aerospace University, Bangladesh Jul 2024–Jun 2025
Design & Development of a Scientific Exploration and Environmental Analysis Module for Assessing Interplanetary Habitability.

Aviation and Aerospace University, Bangladesh Oct 2023–Jun 2024
Design and Development of an Indigenous Mars Rover for Inter-planetary Scientific Exploration.

Awards

Global Finalist — CanSat Competition 2025 in Virginia, USA 2025
Global Finalist — Anatolian Rover Challenge 2025 finals qualification in Turkey 2025
National Champion — Dr. Muhammad Shahidullah Memorial Debate Festival, Dhaka University 2022
Winner — ATN Bangla Television Public Parliament 2022
National Champion — BTV Chattogram Inter College National Debate Competition 2021
National Champion — BTV Chattogram Inter School National Debate Competition 2019
Best Science Project Award — Khulna Divisional Science & Project Fair 2019

National Finalist — Bangladesh Physics Olympiad, Junior Category 2018
National Finalist — Bangladesh Math Olympiad, Junior Category 2018

Languages Proficiency

English **GRE Score**: «000» (*Quant «000»*), **IELTS Score**: «0.0»
Bangla (*Native*)

References

Name	**Employer**	**Title**	**Email**
Air Cdre Dr. Mohammad Akhter Jamil (Thesis Supervisor)	Aviation and Aerospace University, Bangladesh	Dean	jamil8703@gmail.com
Air Cdre (Retd) Dr. Md Afzal Hossain (Project Supervisor)	Aviation and Aerospace University, Bangladesh	Distinguished Professor	afzal@aaub.edu.bd
Dr. Md-Mizanur Rahoman (Research Supervisor)	Begum Rokeya University Rangpur, Bangladesh	Professor	mizan@brur.ac.bd
Dr. S.M. Sadakatul Bari (Course Instructor)	Aviation and Aerospace University, Bangladesh	Chairman	smsbari@aaub.edu.bd
Md Samin Rahman (Course Instructor)	Aviation and Aerospace University, Bangladesh	Assistant Professor	samin@aaub.edu.bd